

PHYS 427: Lecture # 11

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September 30, 2026

Lecture Span

1. Gibbs Free Energy

Exam: 50 mins. also, Z , F , and $\langle U \rangle$. Cheat sheet front and back.

1 Gibbs Free Energy

$$G = U + pV - TS$$

Now, Gibbs free energy had fixed pressure and constant particles. Also, reservoir and the system is closed. Then, the $dS_{R+S} = 0$. Then,

$$dS_{R+S} = dS_R + dS_S = \frac{dU_R}{T_R} + \frac{p_R}{T_R} dV_R + dS_S \quad (1)$$

Also, assume that

$$dU_R = -dU_S \quad (2)$$

$$dV_R = -dV_S \quad (3)$$

$$T_R = T_S \quad (4)$$

$$p_R = p_S \quad (5)$$

Then,

$$dS_{R+S} = -\frac{1}{T} d(U_S + pV_S - TS_S) \iff -\frac{1}{T} dG \quad (6)$$

Therefore, S is maximized if G is minimized.

In a microcanonical ensemble, S is closed with fixed U, V . Entropy can only increase.

In a canonical ensemble, a system and a reservoir are coupled. $T_{R/S}, V_S$ are fixed. Then, F decreases with $\Delta F_S \leq 0$.

In a different ensemble where the system is inside the reservoir and volume can change, $T_{S/R}$ are the same and fixed, $p_{S/R}$ are the same and fixed. Then G decreases and $\Delta G_S < 0$.

$$dU = TdS - pdV + \mu dN \quad (7)$$

$$dF = -pdV - SdT + \mu dN \quad (8)$$

$$dG = -SdT + Vdp + \mu dN \quad (9)$$

For a monoatomic ideal gas,

$$G = U + pV - TS \quad (10)$$

then

$$G = \frac{3}{2}Nk_B T - Nk_B T - Nk_B T (\ln(n_Q/n) + 5/2) \quad (11)$$

$$= Nk_B T \ln(n/n_Q) \quad (12)$$

Also, μ for an ideal gas

$$\mu = k_B T \ln(n/n_Q) \implies G = N\mu \quad (13)$$

Such a relationship is true in general. In general,

$$G = \sum \mu_i N_i \quad (14)$$

Example: Chemical equilibrium

For chemical reactions under fixed T, p . Then, G is the thermodynamic potential. In equilibrium, G is minimized. Nitrogen fixation:



Or

$$2NH_3 - N_2 - 3H_2 = 0 \quad (16)$$

By convention, reactants are negative and products are positive. In equilibrium,

$$dG = 0 = -SdT + Vdp + \sum \mu_i dN_i \quad (17)$$

Then, dN_i is the changes in the # of molecules for species i during the reaction. At fixed pressure and temperatures,

$$dG = 0 = \sum_i \mu_i dN_i = \mu_{N_2}(-1) + \mu_{H_2}(-3) + \mu_{NH_3}(2) \quad (18)$$

Then,

$$0 = -\mu_{N_2} - 3\mu_{H_2} + 2\mu_{NH_3} \quad (19)$$

This technique for equilibrium condition can be generalized. We can use the ideal gas formula to approximate this reaction: $p_i = n_i k_B T$ where $n_i = N_i/V$. For an ideal gas,

$$\mu_i = k_B T \ln\left(\frac{n_i}{n_Q}\right) = k_B T \ln\left(\frac{p_i}{n_Q k_B T}\right) \quad (20)$$

People can also write it as

$$\mu_i = \mu_i^0 + k_B T \ln(p_i/p_0) \quad (21)$$

p_i is the partial pressure which is hypothetical pressure that gas i would have if it alone occupied the volume. Then, p_0 is 1 atmosphere. And

$$\mu_i^0 = k_B T \ln\left(\frac{p_0}{n_Q k_B T}\right) \quad (22)$$

We can then rewrite everything as

$$\frac{p_{NH_3}^2 p_0^2}{p_{N_2}^2 p_{H_2}^3} = \exp\left(-\frac{\Delta G^0}{k_B T}\right), \quad \Delta G^0 = -\mu_{N_2}^0 - \dots \quad (23)$$

This is the law of mass action. In dilute solutions, the same steps work since you can write chemical potentials as

$$\mu_i = m_i^0 + k_B T \ln(n_i/n_0) \quad (24)$$

Where n_0 is 1 mol/liter and n_i is the concentration of solute given in mol/lit.